

Putting It All Together

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We've reached our final chapter describing the science behind high-intensity interval training (HIIT). Understanding the science is important for a practitioner because it instills us with the power to be able to manipulate the contents of training to form and fine-tune our HIIT weapons. If we have the right intelligence behind our weapons, we can operate more like Rambo, moving through the battlefield more astutely, hitting our desired physiological response targets, as opposed to having nothing but shock-and-awe-type operations in our repertoire, which leave nothing but collateral damage in their wake. Taking the targeted approach to HIIT application we have outlined provides our solutions to the programming puzzle.

In this chapter, we take the opportunity to briefly review each chapter (figure 10.1), simplify the key points, and bring them all together before delving into their specific application in the 20 most popular individual and team sports in the world that use prescribed HIIT as a key preparation tool. Here, you will read about how practitioners immersed in your favorite sport, or even those outside of your current interest perhaps, apply the science of HIIT within the confines of their specific world—elite or high-performance sport.

Chapter 1: Genesis and Evolution of High-Intensity Interval Training

In chapter 1, we started with some background in the area, describing the origin, history, and evolution of HIIT, before arriving at our current state of understanding its best practice application today. HIIT is simply defined as repeated bouts of high-intensity exercise performed above the lactate threshold (a perceived effort of hard), interspersed with either low-intensity exercise or complete rest. Ultimately, this broken exercise format of HIIT can allow either a more manageable training session with an equal training stress, or alternatively, the accumulation of a greater total volume of high-intensity exercise when the sequence is repeated. By performing a portion of our training in the HIIT format, as opposed to continuous exercise at an equivalent average work rate only, we can apply a better global training stimulus, alongside adaptations that can be made specific to a given biological system and/or sport demands.

From its beginnings in the early 1900s, long-distance runners were documented performing high-intensity repetitions on the track prior to winning Olympic medals, but as described in the

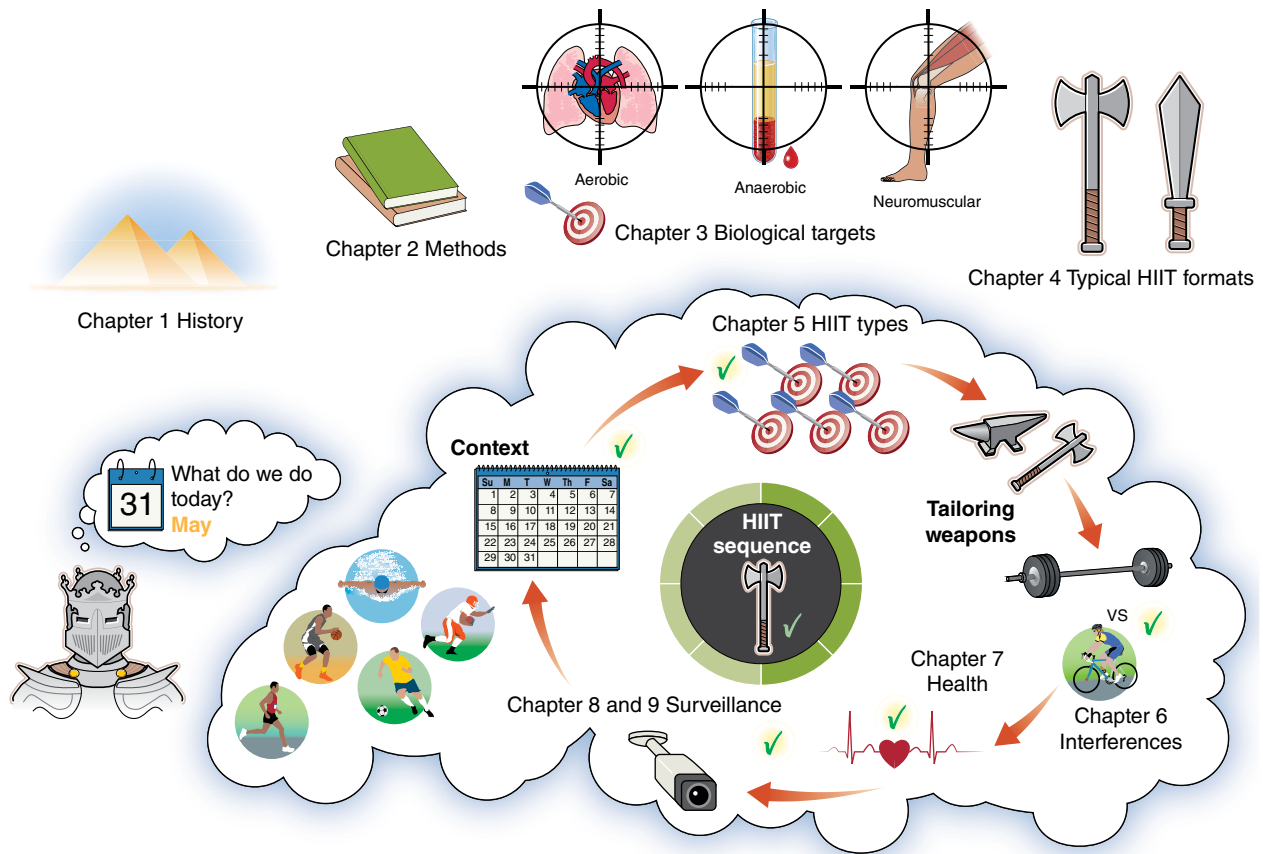


Figure 10.1 Putting it all together. A synopsis of the science and application of the high-intensity interval training journey we have taken in our book.

middle-distance running chapter by J. C. Vollmer (chapter 13), it wasn't until the 1950s when physiologists began describing the response in scientific journals. Team sports took hold of the method in the 1970s, where successful soccer coaches were witnessed using various forms of HIIT to prepare players. The computer, Internet, television, and other technological advances in media helped facilitate the global rise in sport interest over the last five decades into the multibillion-dollar industry it is today. As appropriately applied HIIT is considered a key preparatory tool in reaching individual athlete and team goals, its scientific interest has grown in parallel, where today it now ranks as the highest interest topic by leading organizations in sports science and sports medicine (15).

One of the primary aims of the book was to show the reader how HIIT can be used as a tool or weapon to solve a practitioner's programming puzzle. Indeed, by manipulating the different features of HIIT, we can create various types of weapons through their associated metabolic and neuromuscular responses,

alongside the resulting adaptations that ensue. But before we dive into the intricacies of making weapons, if we are to actually have an impact in our sport of interest, we need to be very clear about certain aspects of our sport—that is, sport context, and specifically just how appropriate, or not, it might be to launch such military weapon engagements (figure 1.4). While we are fortunate to have the research available to understand and describe our tools (5, 6), such information can be used effectively only when we clearly understand the real-world operations of our athletes inside the sport of interest (3). HIIT needs to be applied at the right place and the right time. To this effect, the primary figure used throughout the text, figure 1.5, provides a global synopsis of the situation and highlights how we must understand the individual athlete and his or her physiological profile, the phase of training the athlete is in (micro-, meso-, macrocycle), and the athlete's needs relative to the sport demands (figure 1.6), prior to selecting the appropriate HIIT session to use. Once we clearly see the features of the battlefield before

us, only then can we send in our Navy Seal team to hit the precise physiological target responses needed to generate the adaptations we are after. What we are trying to get away from, and where we feel some may be doing their sport a disservice, is when we enthusiastically take the shock-and-awe approach with its associated collateral damage and uncertain outcomes for athletes. Indeed, it would be best if the term “no pain, no gain” were removed from our vernacular.

Chapter 2: Traditional Methods of HIIT Programming

We used chapter 2 to step back and understand the lay of the land, so to speak, in the world of HIIT science and application, looking at traditional methods of HIIT programming. Certainly there are many useful concepts in the area we need to appreciate, which have been developed over time from leading coaches and practitioners across the years, methods we still use today. Techniques included the rating of perceived exertion technique, the maximal aerobic speed and/or maximal aerobic power-based methods, the 30-15 intermittent fitness test (30-15 IFT), the use of proportions of the anaerobic speed reserve, all-out sprint training, the track-and-field or team sport approaches, and also heart rate- and power meter-based approaches. All approaches are considered useful in the right context; all have their pros and cons, and none are perfect. However, it is important to have different tools in the toolbox to apply what's best at the right place, at the right time. We will not elaborate further on these methods here, but please return to review chapter 2 as necessary for details related to any of these traditional methods (6).

Chapter 3: Physiological Targets of HIIT

In chapter 3, we began the process of describing how different forms of HIIT can be used to alter physiological responses. Notwithstanding the multitude of biochemical and physiological aspects that are impacted during various forms of HIIT (11), we tried to follow Einstein's advice, when he said famously, “things should be made as simple as possible, but no simpler.” Here, we expanded on the detail we started in our two-part review (5, 6), by describing the key physiological aspects that are affected with HIIT. These included:

- Aerobic oxidative metabolic response targets, including cardiac output, aerobic enzymes, mitochondrial function as well as respiratory work;
- Anaerobic glycolytic metabolic response targets that may cause changes primarily in energetic flux, lactate kinetics, and glycogen depletion; and
- Neuromuscular and musculoskeletal strain targets seen through various aspects including soreness, muscle function and muscle strength or power, and changes in movement characteristics.

The manipulation of the different HIIT parameters, outlined in chapter 4, can form our HIIT target types (figure 1.5) that we can fine-tune further from knowledge gained in chapter 5.

Chapter 4: Manipulating HIIT Variables

In chapter 4, we used a strange English saying that “there's more than one way to skin a cat” when it comes to HIIT, before introducing the 12 different factors that can be manipulated to form our primary HIIT weapons. While the main factors were the intensity and duration of the work bouts, other variables included the intensity and duration of the recovery interval, the number of interval bouts or series duration, the number of interval bout series, their between-series recovery duration and intensity, the total work performed, the exercise mode and/or ground surface used for run-based HIIT, the environment (heat and altitude), and nutritional elements manipulated around the HIIT session.

The chapter highlighted that, depending on how you skin the cat, in general, there were five main classifications of HIIT weapons we can form, including:

1. Long intervals, that include repeated bouts of exercise around $v/p\dot{V}O_{2\max}$ (95%-105%) or 80% to 90% of V_{IFT} , involving durations longer than a minute, separated by short durations (1-3 min) of passive recovery, or longer durations (2-4 min) of active recovery up to 45% of V_{IFT} or 60% of $V/P_{IncTest}$.
2. Short intervals, consisting of interval bouts performed at between 90% and 105% of V_{IFT} (100%-120% $V/P_{IncTest}$) for less than 60 s, and

repeated in comparable short-time-duration recovery periods that range in intensity from passive to 45% $\dot{V}_{IFT}$ /60% $\dot{V}/P_{IncTest}$.

3. Repeated-sprint training (RST), which involves 3-10 s all-out efforts with variable recovery durations (majority <30-45 s) ranging from short and passive to 45% $\dot{V}_{IFT}$ /60% $\dot{V}/P_{IncTest}$.
4. Sprint interval training (SIT), consisting of all-out maximal sprinting efforts, ranging from 20 to 30 s. Recovery for such intervals is passive and generally long (typically 1-4 min).
5. Last but not least, we have game-based HIIT (GBHIIT), which includes small-sided games that ultimately wind up forming game-based forms of long intervals that are sport specific. Importantly, GBHIIT involves decision-making and interactions with opponents (at least one in the case of racket sports) and teammates, which make these unique and different from typical sport-specific intervals (often in the form of HIIT with short intervals with athletes reproducing sport-specific movements). GBHIIT usually involves intervals that last 2 to 4 min at a sport-specific intensity of effort, with recovery durations of 90 s to 4 min that are typically passive.

Figure 10.2 offers a summary of the intricacies pertaining to the five key HIIT weapons and their associated targets.

Chapter 5: Using HIIT Weapons

After we established our five primary weapons from chapter 4, we shifted focus in chapter 5 and began to consider how these weapons could be manipulated to hit the primary physiological targets of importance discussed in chapter 3. Depending on how we skinned the cat, HIIT manipulations could allow us to form five different target types, depending on the degree of oxidative, glycolytic, and neuromuscular emphasis the manipulation evoked (figure 10.2). Remember here that the HIIT formats or weapons (long intervals, short intervals, RST, SIT, GBHIIT; figure 10.2) are different than the physiological targets (chapter 3) that we're aiming to hit. The HIIT target types relate to the combination of physiological responses they elicit (degree of oxidative, glycolytic, or neuromuscular response; figure 10.2). Importantly, format does not obligatorily imply target type and different weapons can be

used to hit the same, or alternate, targets. The informed decision is up to you, the four-star practitioner (figure 10.1).

To recap, we suggested that there were primarily six key targets we can aim for to enhance the related systems using our five basic weapons (figure 5.1). These were outlined for us originally in figure 1.5 and included:

- Type 1, metabolic, but eliciting essentially large requirements from the cardiopulmonary system and oxidative muscle fibers;
- Type 2, metabolic as per type 1, but with a certain degree of neuromuscular strain;
- Type 3, metabolic as per type 1, but with a large degree of anaerobic glycolytic energy contribution;
- Type 4, metabolic as per type 3, plus a certain degree of neuromuscular strain;
- Type 5, metabolic, with essentially an important anaerobic glycolytic energy contribution and a large neuromuscular strain;
- Type 6, not considered HIIT, but exercise involving a high neuromuscular strain only, referring typically to speed and strength training.

To manipulate the acute responses to HIIT in terms of cardiopulmonary, anaerobic, and neuromuscular strain, and specifically so as to reach the different targets, the following programming aspects should be considered.

Manipulate the Cardiopulmonary Responses

Most HIIT formats, if properly manipulated, can enable athletes to reach $\dot{V}O_{2max}$ while achieving a prolonged T at $\dot{V}O_{2max}$. However, RSS and SIT sessions allow limited T at $\dot{V}O_{2max}$ compared to HIIT sessions involving long and short intervals. The $\dot{V}O_2$ responses during RSS and SIT appear to be fitness dependent, with fitter athletes less able to reach $\dot{V}O_{2max}$ during such training. Overall, the following general recommendations can be made:

- To individualize exercise intensity and target specific acute physiological responses, $v/p\dot{V}O_{2max}$ and ASR/APR or $\dot{V}_{IFT}$ are likely the more accurate references needed to design HIIT with long (>1-2 min) and short (<45 s) intervals, respectively. For run-based HIIT sessions, compared to the ASR, $\dot{V}_{IFT}$ integrates

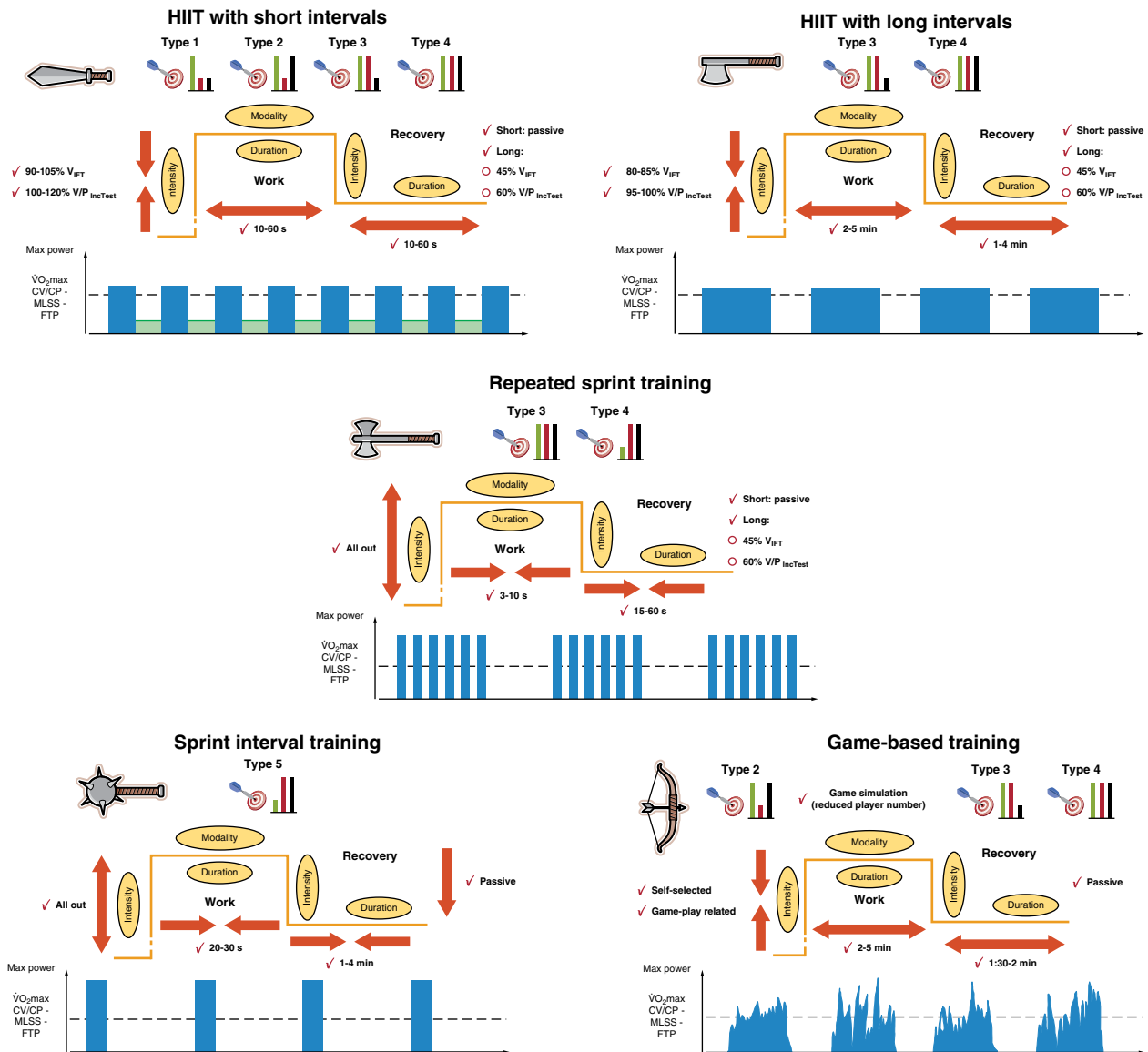


Figure 10.2 Details of the five key HIIT weapons and the associated targets they can hit from their manipulations. See text for details, and additionally chapter 4.

between-effort recovery abilities and COD capacities, which make $\dot{V}_{IFT}$ especially relevant for programming short, supramaximal intermittent runs performed with COD, as implemented in the majority of team and racket sports (see chapter 2).

- Especially in well-trained athletes who perform exercises involving large muscle groups, and assuming the accumulation of T at $\dot{V}O_{2max}$ may maximize the training stimulus to improve performance potential, we recommend long- and short-bout HIIT with a work:relief ratio >1.

Additionally there should be little delay between the warm-up and the start of the HIIT session so that T to $\dot{V}O_{2max}$ is accelerated. Warm-up intensity can be $\leq 60\%$ to 70% $\dot{v}/p\dot{V}O_{2max}$ or game-based (moderate intensity) for team and racket sport athletes. Combined data from high-level athletes suggest that long intervals and/or short intervals with a work:relief ratio >1 should enable a greater T at $\dot{V}O_{2max}$ /exercise time ratio during HIIT sessions. Total session volume should enable athletes to spend between ≈ 5 min (team and racket sports) and ≈ 10 min (endurance sports) at $\dot{V}O_{2max}$.

Manipulate the Anaerobic Contribution

In practice, some HIIT formats are associated with a large anaerobic glycolytic energy contribution. It is possible to minimize the anaerobic system participation by using certain forms of HIIT, including short intervals, and possibly some types of GBHIIT. When appropriately manipulated, HIIT sessions (especially RSS or SIT) can, in contrast, be a powerful stimulus for producing high levels of blood lactate, a marker of the anaerobic glycolytic response.

Manipulate the Neuromuscular Load and Musculoskeletal Strain

In practice, while the magnitude of neuromuscular load during HIIT can be modulated through the manipulation of HIIT variables (e.g., work intensity or duration, exercise mode or pattern), the responses are highly athlete profile dependent, with endurance-type athletes showing low levels of acute fatigue and speed decrement, and team sport athletes typically showing high levels of neuromuscular fatigue following HIIT. There is likely a bell-shaped relationship between exercise intensity and acute neuromuscular performance responses, with too low ($\leq 85\%$ $\dot{V}O_{2\max}$) and too high (all-out) an intensity having not enough and acute detrimental effects, respectively. Practitioners can choose and balance the level of neuromuscular engagement associated with a given HIIT format, based on both the expected training-induced adaptations (either through the HIIT session itself or the associated sessions and possible additive effects) and the acute changes in neuromuscular performance. Running pattern (e.g., COD, introduction of jumps during the recovery periods), exercise mode (e.g., cycling, running, bouncing), or ground surfaces (e.g., pavement, synthetic track, grass, sand, treadmill) and terrain (uphill, downhill) also may have direct implications on traumatic and overuse injury risk, and should be chosen for programming based on a risk/benefit approach.

Chapter 6: Incorporating HIIT Into a Concurrent Training Program

In chapter 6, Jackson Fyfe took us through the topic of concurrent training, defined as the simultaneous integration of both strength and metabolic training

within a periodized training regime, which represents a clear challenge for the practitioner's programming puzzle, with particular appreciation for the type 6 target and response (figure 1.5). While concurrent training has the potential to compromise aspects of strength training adaptation, current evidence suggests it does not attenuate improvements in factors such as $\dot{V}O_{2\max}$ and endurance performance, and in general appears to mostly benefit running and cycling performance. Thus, the main programming puzzle challenge for the practitioner is to manipulate HIIT variables (chapter 4 and 5) so that any interference effect to strength training adaptations is minimized, while simultaneously maximizing improvements in cardiorespiratory fitness via metabolic conditioning. When strength training is performed alone (i.e., single-mode strength training; type 6 target), performance is maximized, as residual neuromuscular fatigue should be minimal, thereby allowing for an optimal adaptation stimulus for strength, muscle hypertrophy, and power development. The practitioner should consider that a less-fatigued athlete can likely:

- lift heavier absolute loads (at a higher percentage of 1RM) for a given number of repetitions,
- perform more repetitions at a given percentage of the 1RM load, and/or
- perform repetitions at faster intended and/or actual concentric velocities (as a percentage of the maximum velocity possible with a given load).

The metabolic conditioning (i.e., endurance and HIIT sessions) component of concurrent training however, likely causes residual neuromuscular fatigue that may compromise both strength training performance and potentially the adaptive response to strength training sessions. Both of these factors may limit the stimulus for adaptation, resulting in attenuated development of classic strength training adaptations.

In the context of prescribing HIIT in concurrent training programs, particularly important considerations include the neuromuscular demands of individual HIIT sessions, and how these demands may then influence performance during subsequent strength training sessions (figure 1.5). Any HIIT session that evokes a large neuromuscular strain component (i.e., type 2, 4, 5) will likely result in more substantial residual fatigue of the exercised musculature, compared to HIIT sessions with smaller

neuromuscular strain components (i.e., type 1, 3). Similarly, it is likely that strength sessions (type 6) may also compromise performance during subsequent HIIT sessions with high neuromuscular demands (i.e., type 2, 4, 5). Thus, practitioners should consider the relative priority of key HIIT or strength sessions within any given training week and manipulate the organization of these sessions to minimize the potential for negative interactions between training modes.

Here are our general recommendations for limiting interference to strength training adaptations with concurrent training.

Limit neuromuscular fatigue prior to strength training sessions by performing strength training either before, or at least (approximately) 6 h after, HIIT sessions.

HIIT sessions performed in the 24 h prior to strength training sessions ideally should be characterized by lower neuromuscular demands. This can be achieved by several tactics:

- Incorporate lower absolute/relative work intensities that are associated with lower neuromuscular load (i.e., type 1 and 3). It is however worth noting that using lower HIIT intensities may often need to be associated with longer effort durations (e.g., 3 min at 85% V_{IFT} versus 10 s/20 s at 120% V_{IFT}), and consequently these longer durations may also cause neuromuscular fatigue.
- Limit prescription of all-out HIIT modalities (i.e., RSS, SIT formats, or in general, neuromuscular-demanding HIIT formats, i.e., type 2, 4, and 5).
- Perform running-based HIIT on softer surfaces (e.g., grass or sand instead of road) likely to impart less neuromuscular strain.
- Limit the change-of-direction (COD) component (particularly sharp COD) of running-based HIIT.
- Employ cycling-based, rather than running-based, HIIT (depending on the athlete's training specificity).

Where possible, and where individual strength/power development is favored over the metabolic conditioning component, total weekly metabolic conditioning volume should be limited by:

- reducing the weekly frequency of HIIT sessions and/or performing fewer repetitions and/or series during HIIT sessions, and

- reducing training volume associated with other cardiorespiratory modalities (e.g., lowering volume of long duration aerobic conditioning sessions).

Chapter 7: HIIT and Its Influence on Stress, Fatigue, and Athlete Health

In chapter 7, Philip Maffetone showed us a new perspective on HIIT that needs to be contemplated when we're considering our athletes. Exercise is a type of stress, and HIIT can represent one of its most potent forms. In this chapter, our aim was to present more of a holistic understanding of stress before more objectively attempting to specifically quantify the stress associated with exercise training (load; chapter 8) as well as the individual response to that training load that arises in the athlete (chapter 9). Thus, chapter 7 was devoted to understanding not only the stress of HIIT but the stress of other confounding variables, including food, that can impact an athlete's physiology, attenuate or compound overall stress, and subsequently improve or derail progress.

In this regard, it was important first to define health and fitness separately. *Fitness* was defined as the quality of human performance and the physical ability to compete at one's athletic potential. *Health* was defined separately more as a state of physical, biochemical, and mental-emotional well-being in which all bodily systems (metabolic, neuromuscular, hormonal, immune, digestive, etc.) function in harmony. Stress does not necessarily need to be considered in a negative light, as it is important, creating the stimulus that enables the body to adapt to the wear and tear of lifestyle, including sport and exercise. However, whether training involves too much workout volume or intensity (load; chapter 8), inadequate nutrition, or poor sleep or recovery, excess stress can impair health and ultimately fitness. Too much stress without adequate recovery quickly manifests in a lack of health, seen as illness, injury, or both. A holistic perspective on HIIT can be appreciated when we consider it across other physiological systems that are affected by it. The hypothalamic-pituitary-adrenal (HPA) axis, which serves as a primary-link between the nervous and endocrine systems, was described as a key regulator of the stress response, and one that is important to appreciate when stress reaches chronic levels that may manifest

in what is more commonly known as the overtraining syndrome.

Sleep and adequate nutrition were singled out as the critical factors of importance for eliciting repair and adaptation to stress. A person's diet is considered the sum of energy (calories) and nutrients obtained from foods and beverages consumed regularly, and requires adequate levels of macronutrients, micronutrients, and phytonutrients. A low-quality, poor diet is one that includes more processed, packaged, and preprepared foods that have lower nutrient density (such as micronutrients and fiber) versus natural, whole, unprocessed foods higher in nutrient density. In addition, processed (refined) carbohydrates, often termed junk food, including flour- and sugar-containing foods, substantially increase the glycemic index of meals and snacks, and can contribute to chronic inflammation and stress through the HPA axis. Indeed, the combination of excess training stress, including HIIT, can create a neuroendocrine imbalance, which will be manifested by the consumption of a poor diet. A diet high in sugar can promote inflammation, metabolic substrate imbalance (reduced fat oxidation), maladaptation, and contribute to physical, biochemical, and mental-emotional injuries. Athletes, coaches, and clinicians need to carefully monitor training progress with sound surveillance methods (chapters 8 and 9) and rely on healthy eating to maximize outcomes from HIIT.

Chapter 8: Quantifying Training Load

In chapter 8 we learned how to put surveillance systems over the markers that matter and monitor them. The primary variable of interest when it comes to HIIT stress and trying to quantify it is the theoretical concept of training load, where training load is defined as the product of training intensity and training volume. These training load parameters can be considered from either an external (occurring outside of the person, e.g., power output or movement) or internal (measured as something going on inside the person, e.g., heart rate during exercise) standpoint. External training load parameters are often used for programming HIIT (i.e., perform 4×3 min at 350 W cycling, 3 min rest; or 4×3 min at 3:15/km running, 2 min rest), while the internal training load refers to the internal training stress of the measured system (heart rate, lactate, etc.) generated during the exercise. Understanding both is important, as either parameter

can be useful to appreciate in light of the other. However we want to skin it (using internal or external loads), we appreciate that these loads represent the likely input marker of stress on the body. This load is required both for programming as well as for understanding what has actually occurred in training (prescribed versus actual training).

Nowadays, we have a surplus of potential technology and its associated data available to consider as being useful. External training load markers should be specific to the athlete's sport and could include power meters, GPS, local positional systems, and semiautomatic video. Relevant internal training load parameters involve those that offer surveillance over the oxidative, glycolytic, and neuromuscular parameters, and might include heart rate, lactate, and rating of perceived exertion.

Finally, in the spirit of Occam's razor, where the simplest explanation tends to be the right one, it can be useful to break down training and HIIT into its relevant intensity brackets by describing training in terms of time accumulated in certain training-intensity bandwidths or zones. Thus, speed, power, and heart rate zone descriptions of athlete training have become useful methods of providing surveillance over athlete training. In the case of the field-based practitioner, the simple and most relevant markers to monitor for load surveillance include power output and/or locomotor movement (GPS), heart rate, and rating of perceived exertion. While the important glycolytic (lactate) load may be absent from our model, this measurement today is currently too impractical for most field-based scenarios, and we await with interest as to whether technology helps create this missing puzzle piece.

Chapter 9: Response to Load

In chapter 9, we explored aspects of the right side of figure 8.2, with particular attention on the output of our training system, or our *response* to training. As load versus response to load are exclusively different concepts, we need a unique set of tools and understanding. Importantly, our athlete's training load response, commonly termed training status, is not uniform across all athletes, and clearly depends on the training and individual characteristics.

We identified three key families of training load response markers that we need to monitor. First is internal metabolic, cardiopulmonary, and autonomic status with heart rate (HR) the tool chosen to infer

on the status of that system. From a measure as simple as HR, we identified a large number of indices that could inform us globally concerning metabolic, cardiopulmonary, and autonomic fitness. In as little as 1 to 5 min, morning resting HR(V) can capture a snapshot of an athlete's cardiac parasympathetic activity on that day, while a period of submaximal exercise HR (last min during 4-5 min cycling or running) is additionally a useful monitoring variable. Resting HRV measures can be collected frequently to examine acute (daily recordings) and chronic (at least 3-4 d per week) responses to training and are likely sensitive to both positive and negative adaptations. Exercise HR may inform more on chronic and positive adaptations.

Next is neuromuscular status and locomotor efficiency using GPS and accelerometry. While insights into an athlete's neuromuscular status in response to training can be gained via different approaches, we focused exclusively on the methods available to practitioners in the field. The most relevant methods include regular vertical jump testing and more importantly microtechnology-derived indices relating to locomotor efficiency and stride characteristics. The use of those latter variables is still in its infancy, but there is no doubt that the development of these technologies in the future will increase our ability to collect better data at a greater frequency, allowing for an improved monitoring sensitivity.

Finally we turn to overall health, fatigue, and mood (psychology) using wellness questionnaires as a simple means of gaining insight into this important (and global) aspect. The relative simplicity of using wellness measures makes them an easy go-to tool for every practitioner's surveillance toolkit. However, it is important for practitioners to appreciate that this effectiveness depends on establishing strong foundations in data quality and continuing to invest in the process. Incorporating wellness measures can aid in the interpretation of HR and GPS measures. Likewise, these former markers give enhanced context to wellness measures to guide action for the practitioner.

Note here that we don't tend to monitor anaerobic-related responses in parallel with the lack of surveillance around this system in load monitoring (chapter 8).

In summary, we've come full circle now and our feedback loop is complete. You, the four-star coach or practitioner in the operation (figure 10.1), understand your battlefield, are aware of the athlete or team in front of you, know the sport and factors of

importance, and as a result, clearly see the context by which you can now appropriately apply your HIIT. With this knowledge at the forefront, alongside the physiological responses you're after (i.e., the three main physiological targets we have been discussing throughout the book—oxidative, glycolytic, neuromuscular), it's time to start planning. You know the types of weapons you can formulate and the factors you can manipulate to fine-tune them to hit your HIIT target types. You put the plan in place, as well as appreciating concurrent training aspects if relevant. With a holistic appreciation of stress, it's time to appropriately monitor both the load and the response to load using simple surveillance techniques. This response helps you to adjust and repeat the process to continuously improve the probability of enhanced outcomes for your individual athletes or teams. In the following application chapters, written by experienced practitioners embedded in some of the world's most popular sports, we'll learn exactly how they go about doing this, and we hope you'll be able to use this information to help refine your own practice.

Application of HIIT in Sport: Thinking Outside the Box

In the forthcoming chapters, expert practitioners will begin by describing their sports and the factors involved with successful campaigns. They will tell us where and when HIIT is important and additionally where and when its relevance takes a back seat to preparation for performance. The targets of physical performance (oxidative, glycolytic, neuromuscular) will be highlighted, and in the case of team sports, how these targets are often different across the various playing positions. Then it's about taking aim at those targets. Here is where the formation and manipulation of the most relevant HIIT weapons, within the context of each sport, is described. Some practitioners even show how their HIIT programs progress and develop from a longitudinal standpoint, as well as offering strategies for structuring the HIIT program within the confines of the sport's controllable and uncontrollable factors. The load and response to load monitoring tools used within each sport are also shown. Finally, example training programs, some with very fine detail pertaining to HIIT and other aspects, are provided throughout most sport chapters. You'll discover that the depth of information gifted by each practitioner is immense, and we know of little else currently

available providing as rich a content as you'll find offered by our team. While traditionally such information is considered top secret by professional teams throughout the world, this group of practitioners holds a different philosophy on the matter, are selfless, open-minded, and are driven more to share with others. These individuals think outside the box (4), and we would describe them as being archetype 3 individuals (figure 10.3).

We believe that the concept of *thinking outside the box* is extremely important both for practitioners working in applied sport and for academics in universities who are passionate about contributing to sport science. Indeed, we often witness the *sport* aspect of sport science as somewhat lost these days in some organizations. Whether you're an academic or a practitioner embedded in sport, or those who strive to do both, we believe that those who embrace the concept described herein and strive to be individuals that fall into personality archetype 3 will have the most impact in both sport science application and research. These individuals are prepared to break from the pack and do what the others are not prepared to do. It is these developed character traits associated with the archetype 3 personality that allow them to do this.

The three personality archetypes in relation to people's ability and willingness to grow, share knowl-

edge, collaborate, and adopt an open-minded attitude are highlighted in figure 10.3, and are based on the idea that people's mindset travels continually back and forth through an infinity-shaped loop, between more (left) and less (right) comfortable zones. It is where people tend to reside for most of the time that defines their profile and that will allow them to make a substantial difference (or not) (4).

Archetype 1

Archetype 1s may be considered to have a balanced profile, reflective of the mindset of most people. They spend most of their time in their comfort zone, but can, when required and/or when pushed by others, step out briefly to grow and make progress. Most won't stay long in the danger zone, however, either through laziness and/or naively believing that "it will be okay." A strong drive for improvement and/or self-confidence to make archetype 3 is missing. Their chance to have impact in both applied sport and/or research is limited.

Archetype 2

Archetype 2s are not our favorite. These individuals we come across often have been working in high-level positions for a long time, both at the academic (e.g., head of faculty or departments, journal editors)

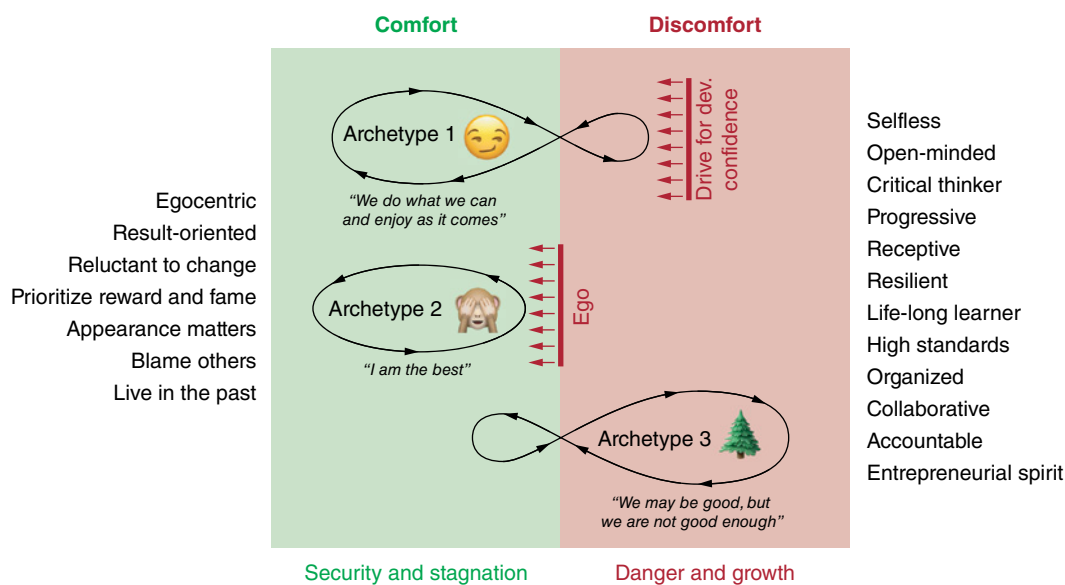


Figure 10.3 Schematic representation of the three personality archetypes. The archetypes are based on the idea that people's mindset travels continually back and forth through an infinity-shaped loop between more (left) and less (right) uncomfortable zones. Drive for dev: drive for development.

Adapted by permission from M. Buchheit, "Outside the Box," *International Journal of Sports Physiology and Performance* 12 no. 8 (2017): 1001-1002.

and practitioner (e.g., head coaches, head physios, and strength coaches) levels. They have chosen to be rather than to do, even though this has meant compromising their own integrity at certain stages of their career. Unfortunately this has little consequence for them, as long as their titles, salaries, and public roles are secured, which satisfies their egos and allows them to feel important. Centered on themselves, they tick the boxes and continue in their same looping comfort zone. They purposely avoid challenge and can't be bothered listening to others. They are more satisfied with old problems than committed to finding new solutions (12). When it comes to giving lectures, they teach what they know and not what the students need. They want their names on papers as last authors, but would be unable to collect the data or discuss the stats and study findings. Their blindness keeps them away from the recent literature and the reality of journal requirements (i.e., topics, quality, designs), which inevitably leads to inappropriate and irrelevant submissions, likely to be rejected by editors. In some extreme scenarios, people keep digging deeper against the evidence. For example, while the need to move away from the null hypothesis testing approach is in the process of being accepted (2, 14), those running in their own circle (or staying in orbit (3)) still ignore effect magnitudes (8) and choose to further decrease the P value threshold to make decisions (<0.005) (1) (see also chapter 9).

In high-performance sports, archetype 2s don't read research papers and never update their skills and methods. They avoid using new technologies and continue delivering outdated programs. They purposely don't share what they do to protect themselves. Their focus on their own personal comfort is so prevalent that it often derails the optimal training or recovery process of the athletes they oversee. "They want to wear the tracksuit but not run the laps" (7). Providing advice to help those individuals is a

waste of time unfortunately, as their ego prevents them from seeing outside the box (figure 10.3). Their future depends on their political and survival strategies.

Archetype 3

Our archetype 3 individuals are featured in the subsequent chapters. These individuals have taken the ultimate progression from the archetype 1 personality toward the right side of the shape (figure 10.3). They are selfless, open-minded, curious, ambitious, and accountable for their actions and show a critical mind. They know well that getting "out of the box" is necessary to learn, grow, innovate, create, and, eventually, succeed. They have understood that life is continuously brought into question and are always willing to do better. They have embraced the uncomfortable truth that natural assets and talent can be outmatched through consistent efforts, deliberate practice (10), and in turn, the development of skills. They are open to constructive criticism (9) and listen more than they talk. They are not afraid to ask for help. They accept and acknowledge their errors to learn from them. They set very high standards for themselves, apply strict self-discipline, and tend to be life-long learners (13). They read daily, listen to podcasts from various fields, travel, seek information in different disciplines, and always say yes when it comes to sharing their experience and knowledge (chapters 11-30). They are more concerned by the process than by the results. They are interested in doing and act to keep their integrity. They treat everyone the same, regardless of their status (13).

For the future sport scientists reading this volume, both in research and as applied practitioners, we feel that getting out of the box is likely a compulsory trait needed to advance our field in the long term. Embrace the challenge, and please continue to advance our field forward!

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